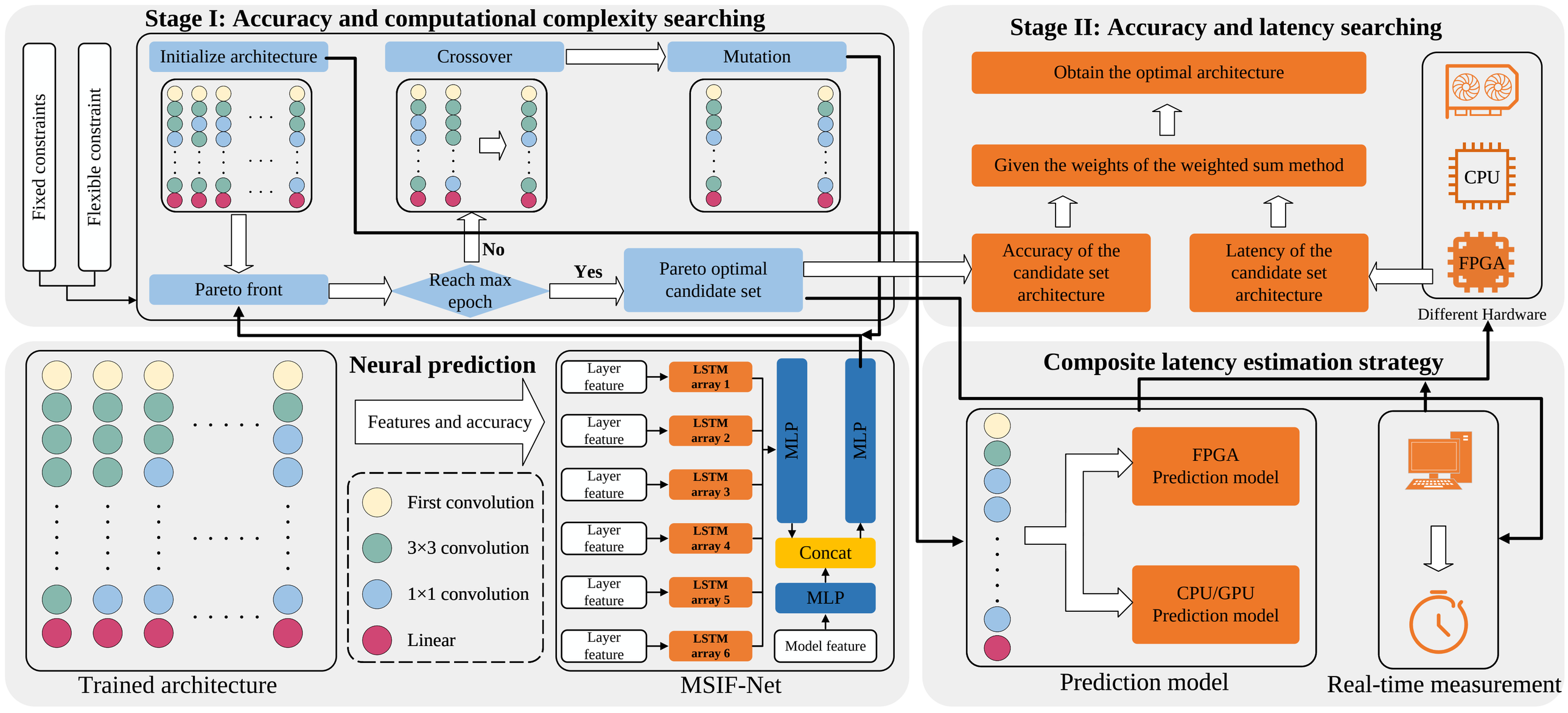
Stage I: Accuracy and computational complexity searching
Fixed constraints
Flexible constraint
Initialize architecture
Crossover
Mutation
Pareto front
Reach max epoch
No
Yes
Pareto optimal candidate set
Stage II: Accuracy and latency searching
Obtain the optimal architecture
Given the weights of the weighted sum method
Accuracy of the candidate set architecture
Latency of the candidate set architecture
CPU
FPGA
Different Hardware
Neural prediction
Features and accuracy
First convolution
3×3 convolution
1×1 convolution
Linear
Layer feature
LSTM array 1
LSTM array 2
LSTM array 3
LSTM array 4
LSTM array 5
LSTM array 6
MLP
MLP
Concat
MLP
Model feature
Composite latency estimation strategy
FPGA Prediction model
CPU/GPU Prediction model
Trained architecture
MSIF-Net
Prediction model
Real-time measurement